IEE575 - Applied Stochastic Operations Research Models
Lab 5

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Roadmap

The Black-Box Optimization Challenge

Key Characteristics

- Objective function $f\colon \mathcal{X} \to \mathbb{R}$ is:
  - Expensive to evaluate
  - Black-box (no derivatives)
  - Possibly noisy observations
- Typical applications:
  - Hyperparameter tuning
  - Materials design
  - Robotics control

Goal

Find

$$x^* = \arg\max_{x \in \mathcal{X}} f(x)$$

where f is costly to evaluate.

- Domain $\mathcal{X}$ is bounded (e.g., $[-3, 3]^d$).
- Observations $y_i = f(x_i) + \epsilon_i$ may be noisy.

Bayesian Optimization Components

Core Components

1. Probabilistic Surrogate Model
2. Acquisition Function
3. Optimization Protocol

Surrogate Model Choices

- Gaussian Processes (Most common)
- Student-t Processes
- Bayesian Neural Networks
- Random Forests

Algorithmic Workflow

1. **Initialization:** Sample initial points $\{(x_i, y_i)\}_{i=1}^{n}$.
2. **Fit GP:** Train Gaussian Process surrogate to model f.
3. **Acquisition Optimization:**

$$x_{\text{next}} = \arg\max_{x \in \mathcal{X}} \alpha(x)$$

4. **Evaluate:** $y_{\text{next}} = f(x_{\text{next}})$.
5. **Update:** Augment data and repeat until budget exhausted.

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BayesianOptimization

```
Given 𝒟_0 = {(x_i, y_i)}_{i=1}^n, acquisition α, budget T
for t = 1 to T do
    Fit GP on 𝒟_{t−1}
    x_t = arg max_x α(x; 𝒟_{t−1})
    y_t = f(x_t)
    𝒟_t = 𝒟_{t−1} ∪ {(x_t, y_t)}
end for
Return best x found
```

Navigating the Search Space

Acquisition Function Roles

- **Exploitation**: Focus on areas with high predicted values
- **Exploration**: Investigate regions with high uncertainty
- **Balance**: Parameterized tradeoff between both

Design Considerations

- Closed-form computability
- Differentiability
- Computational efficiency

Expected Improvement Formulation

$\mathrm{EI(x)} = \mathrm{E}[\max(f(x) - f(x^+), 0)]$
Analytic expression under GP prior:

$$\mathrm{EI}(x) = \begin{cases} (\mu(x) - f(x^+) - \xi)\Phi(Z) + \sigma(x)\phi(Z) & \sigma(x) > 0 \\ 0 & \sigma(x) = 0 \end{cases}$$

where $Z = \frac{\mu(x) - f(x^+) - \xi}{\sigma(x)}$

- ξ: Exploration parameter (default 0.01)
- Φ: Standard Normal CDF
- ϕ: Standard Normal PDF

Practical Implementation Tips

Numerical Stability

- Add jitter to diagonal of covariance matrix
- Use Cholesky decomposition for inversion
- Normalize input space

Hyperparameter Tuning

- Optimize GP hyperparameters via MLE
- Use multiple restarts
- Consider marginal likelihood

Software Tools

- GPyOpt (Python)
- BayesOpt (C++)
- scikit-optimize (Python)
- Dragonfly (Python)

Common Pitfalls

- Poor initialization
- Inadequate exploration
- Overfitting surrogate

Recent Developments in BO

Active Research Areas

- High-dimensional optimization
- Multi-fidelity optimization
- Constrained BO
- Multi-objective BO
- Meta-learning for BO

Recommended Resources

Essential Reading

- **frazier2018tutorial**
- **shahriari2015taking**
- **rasmussen2006gaussian**

Interactive Tutorials

- Distill.pub Visual Guide